Roofline Model
and
Arithmetic Intensity

Introduction

- Slides Courtesy: S. Williams, Xiaoye Sherry Li and others

Slides adapted by: Sparsh Mittal

Roofline Model for a Computing System

- Synthesize communication, computation, and locality into a single metric
- In any program, we need to do two things:
  - Bring data from memory
  - Perform computations on the data
- The time consumed by the program is the higher of the two.
- This intuition is captured by the roofline model.

$$\textbf{Attainable Performance}_{ij} = \min \begin{cases} \textbf{PeakFLOP/s} \\ \textbf{AI * Bandwidth} \end{cases}$$

Definition of Arithmetic Intensity (AI)

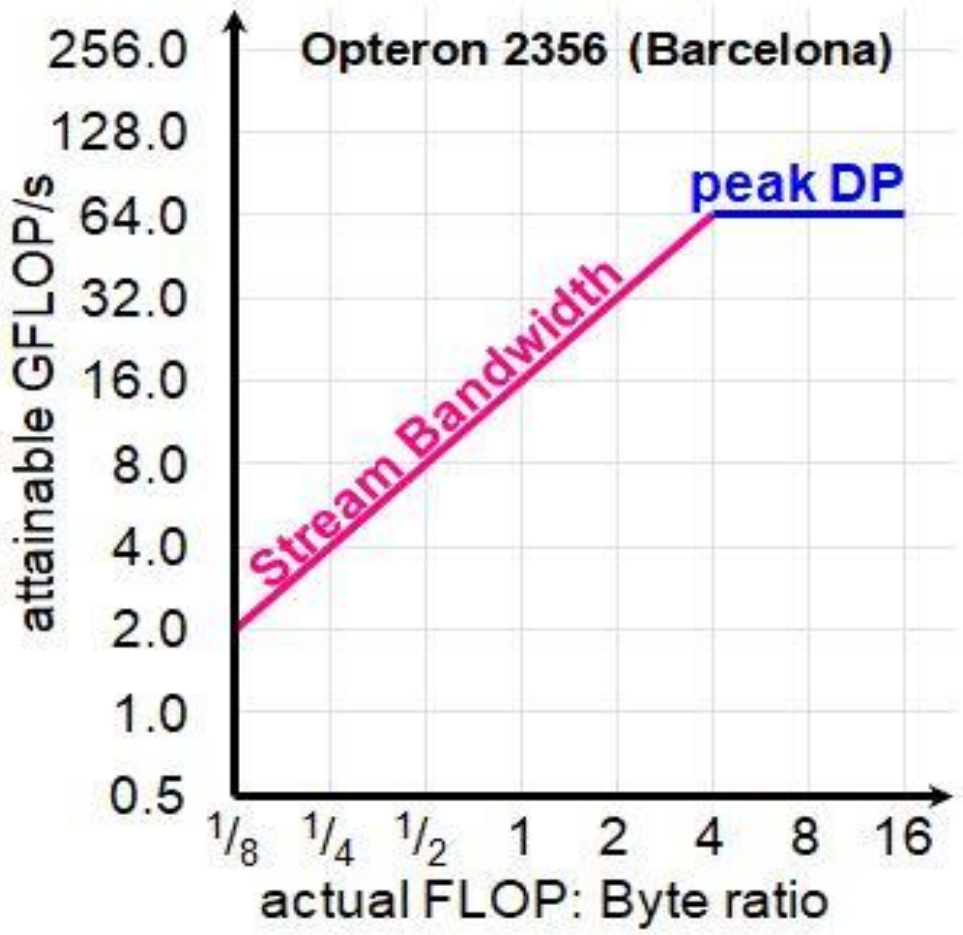
- AI = #Computations/#Bytes Fetched From or stored into Memory
- AI = Total FLOP / Total DRAM Byte
- Roofline model is defined for a system. AI is defined for an application

- Example: consider multiplication of two N*N matrices.
- Number of computations: $O(N^3)$
- Amount of bytes fetched: $O(N^2)$
- AI of matrix multiplication ~ $O(N^3)$/ $O(N^2)$ ~ $O(N)$

An Example Roofline Model

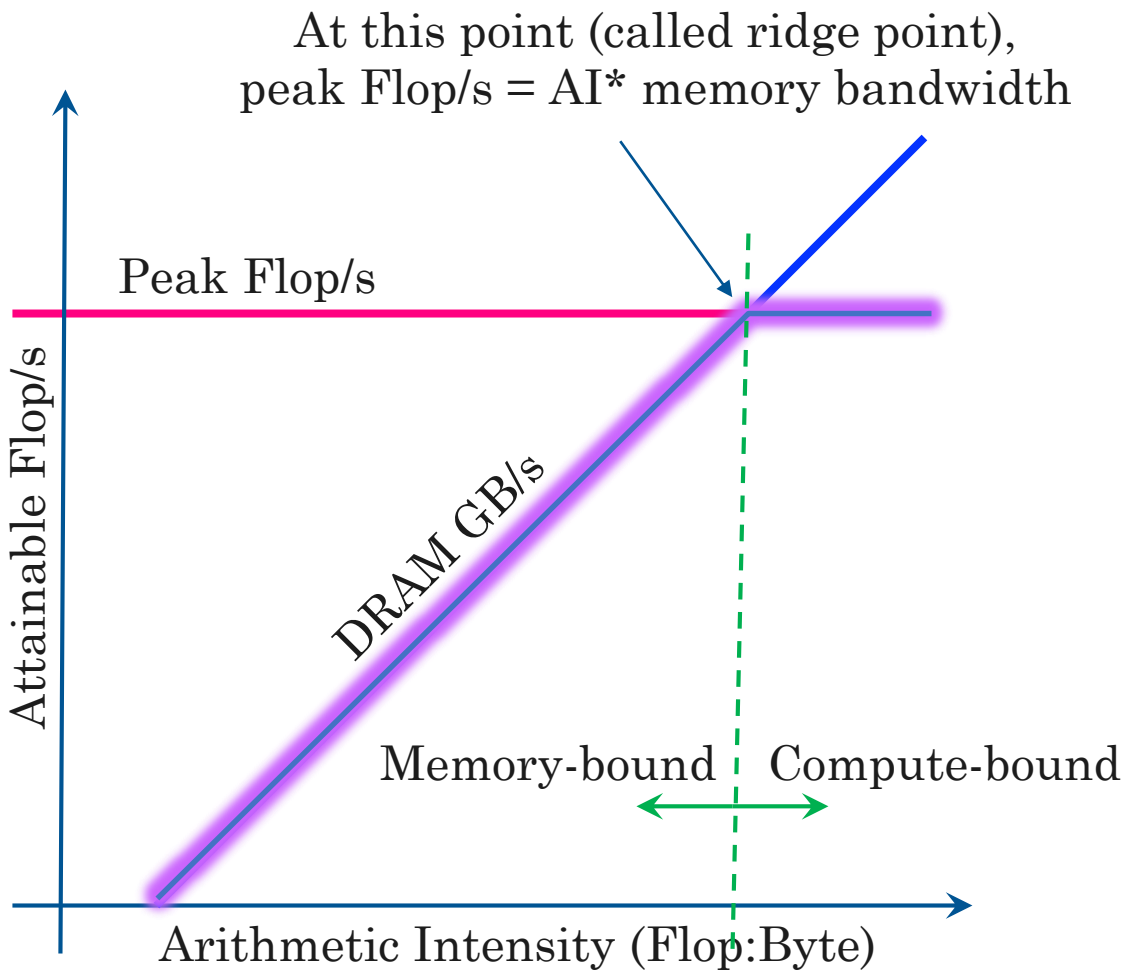
Y-axis is attainable performance

X-axis is AI (FLOP/byte ratio)



DP – Double-precision

Showing AI on Roofline Model



How Roofline Model Is Derived

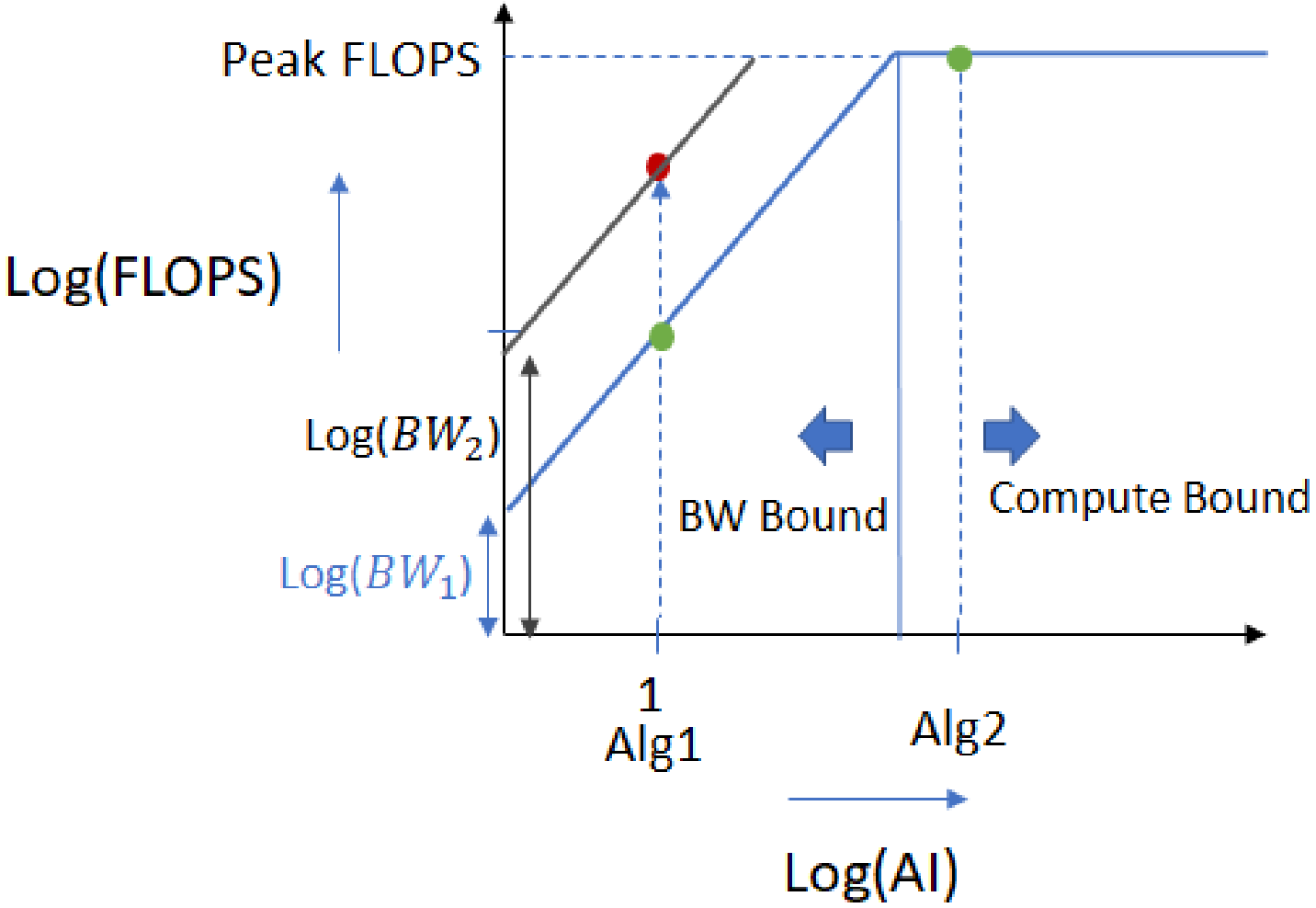
- Bandwidth = byte/second
- FLOPS = FLOP/second
- AI = Total FLOP / Total DRAM Byte
- From above three, we get:
- FLOPS = AI * Bandwidth
- Take log of both the sides:
- log(FLOPS) = log (AI) + log(Bandwidth)

- Roofline model is plot of log(FLOPS) against log(AI)
- y=mx+b, where m=1 and b = log(Bandwidth)

Roofline Model

- For different values of BW, different intercepts on y-axis are obtained

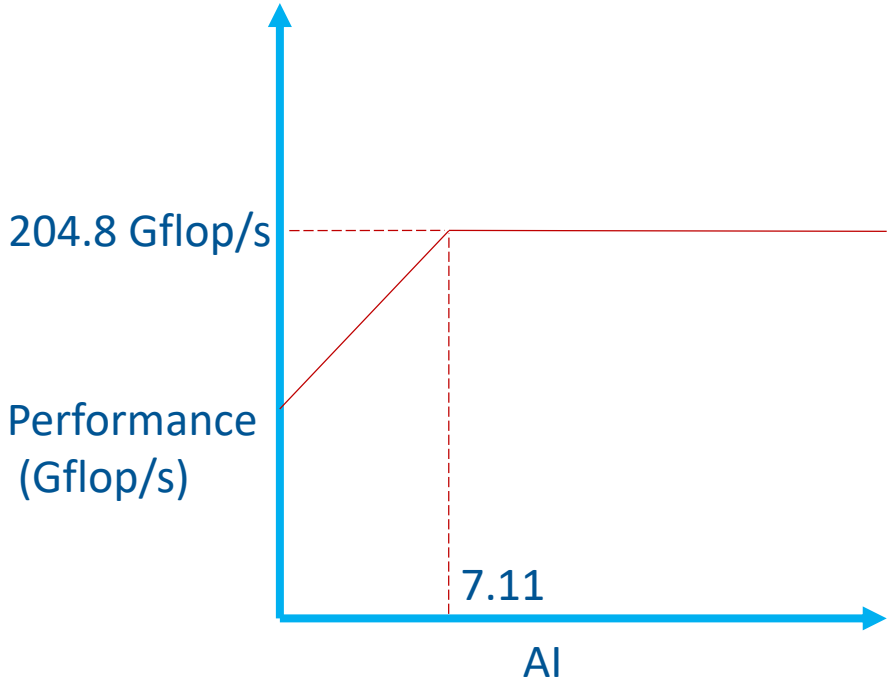
- Different algorithms have different AI value, and hence, they obtain different performances.



BW – bandwidth

Example [1/2]

- Consider a roofline model, whose ridge point has (x, y) coordinates as (7.11,204.8), shown as antilog values. Find the performance for AI = 1/14. Also, tell whether we are in bandwidth-bound or compute-bound region.



Example [2/2]

- Since AI < 7.11, we are in memory bandwidth-bound region.
- At ridge point: AI*bandwidth = PeakFLOP

  7.11* bandwidth = 204.8

  => bandwidth = 28.8
- Now, for AI = 1/14, we get performance as AI* bandwidth = 28.8/14 = 2.057
- With increasing AI, performance is improved, till we reach the peak performance of the processor.

Example: STREAM Triad

```
for(i=0;i<N;i++){
  Z[i] = X[i] + alpha*Y[i];
}
```

- Two (2) flops per iteration
- Each variable takes 8B
- Transfer 24 bytes per iteration (read X[i], Y[i], write Z[i])
- **AI = 2/24 = 0.083 flops per byte**

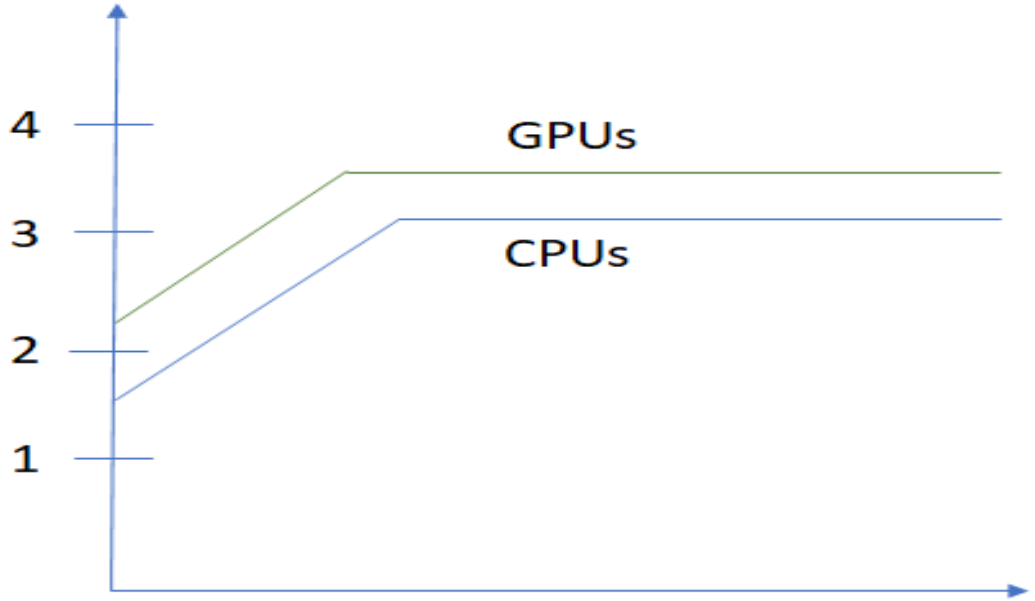
Plotting Roofline Model of GPU and CPU (1/2)

	K80 GPU (single die)	Intel Haswell E5 2699 v3 CPU
Peak GFLOPS (FP32)	2800 GFLOPS(Giga FLOPS)	1300 GFLOPS
Memory BW	160 GB/s	51

- Consider the data above. From this, draw roofline model of CPU and GPU
- Solution: first, compute Log10() value of these numbers.

Plotting Roofline Model of GPU and CPU (2/2)

	K80 GPU (single die)	Intel Haswell E5 2699 v3
Peak GFLOPS (FP32)	2800 GFLOPS Log10(2800)=3.4	1300 GFLOPS Log10(1300) = 3.11
Memory BW	160 GB/s Log10(160) = 2.2	51 Log10(51) = 1.70



#Parameters and Computations in AlexNet

Layer	Number of Parameters	Number of Computations
Conv1	34944	105705600
Conv2	614656	448084224
Conv3	885120	149585280
Conv4	1327488	224345472
Conv5	884992	149563648
FC6	37748736	37748736
FC7	16777216	16777216
FC8	4096000	4096000
Overall	62369152	1135906176
Conv layers	3.7million (6%)	1.08 billion (95%)
FC layers	58.6 million (94%)	58.6 million (5%)

AI of AlexNet (Assuming Each Parameter Is 4B)

Layer	#Parameters	#Computations	AI
Conv1	34944	105705600	756.3
Conv2	614656	448084224	182.3
Conv3	885120	149585280	42.25
Conv4	1327488	224345472	42.25
Conv5	884992	149563648	42.25
FC6	37748736	37748736	0.25
FC7	16777216	16777216	0.25
FC8	4096000	4096000	0.25
Overall	62369152	1135906176	4.55
Conv layers	3.7million (6%)	1.08 billion (95%)	73
FC layers	58.6 million (94%)	58.6 million (5%)	0.25

- AI varies greatly across layers. AI of CONV layers is higher than that of FC layers. CONV layers => compute-bound. FC layers => memory bandwidth-bound.

- Results from a Google's paper
- This figure compares roofline of TPU with that of CPU and GPU

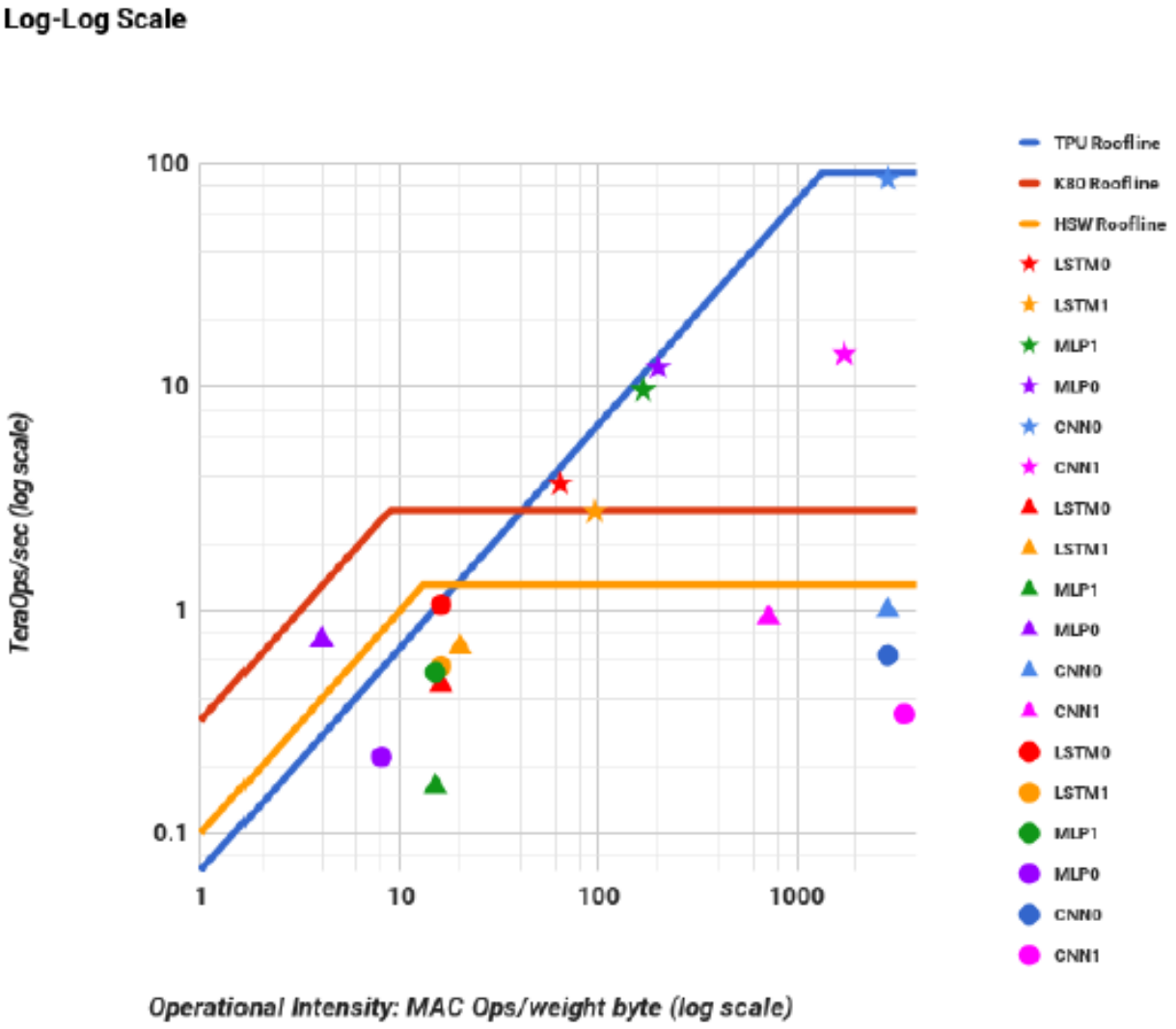


Figure 8. Figures 5-7 combined into a log-log graph. Stars are for the TPU, triangles are for the K80, and circles are for Haswell. All TPU stars are at or above the other two rooflines.

Jouppi, Norman P., et al. "In-datacenter performance analysis of a tensor processing unit." *Proceedings of the 44th annual international symposium on computer architecture*. 2017.

Summary: Key Lessons

- Roofline model and arithmetic intensity
- Finding roofline model/AI of
  - An application
  - Various processors (CPU and GPU)
  - Various layers of a CNN

CNN = Convolutional Neural Network

Further Study

Roofline model was proposed in this paper:

- https://www.researchgate.net/publication/220423225_Roofline_An_Insightful_Visual_Performance_Model_for_Multicore_Architectures

Some solved examples are here:

- https://crd.lbl.gov/assets/Uploads/SC18-Roofline-1-intro.pdf

This is the most famous book on computer architecture, which also discusses about roofline model:

- http://acs.pub.ro/~cpop/SMPA/Computer%20Architecture%20A%20Quantitative%20Approach%20(5th%20edition).pdf.
- https://www.telesens.co/2018/07/26/understanding-roofline-charts/